

# Matthew Woody

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## EDUCATION

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### Sonoma State University

B.S. Applied Mathematics | GPA: 3.41 / 4.00

Expected May 2027

Rohnert Park, CA

- *Relevant Coursework:* Partial Differential Equations, Modern Physics, Numerical Analysis, Mathematical Modeling, Linear Algebra, Data Structures, Probability Theory

### Santa Rosa Junior College

A.S. Mathematics & Natural Sciences

May 2025

Santa Rosa, CA

## RESEARCH EXPERIENCE

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### Brookhaven National Laboratory

Atmospheric Science Intern

June 2026 – August 2026

Upton, NY

*Advisor: Dr. Mindy Deng*

- Integrated NEXRAD Level II polarimetric radar data with minute-resolution CoURAGE surface observations to construct an event-based dataset of 18 cold front passages across heterogeneous terrain and land-cover regimes
- Developed and validated a Python-based radar feature tracking workflow using polarimetric quality control, connected-component filtering, sensitivity-tested thresholds, lag-correlation propagation estimates, and DBSCAN clustering to isolate frontal precipitation
- Retrieved and composited polarimetric radar-derived drop size and rainfall properties across four radar transects to evaluate variability with terrain, season, and Chesapeake Bay breeze occurrence, producing inspectable time-range diagrams and spatial profiles.
- Identified recurring mountain-region enhancement in cold front precipitation with longer duration, broader occurrence of high instantaneous rain rates, and  $\sim 2\text{-}4\times$  greater accumulation than urban region precipitation during the analyzed spring and summer cases
- Evaluated methodological uncertainty and radar sampling limitations including beam geometry, retrieval assumptions, event-to-event variability, and sensitivity to feature selection thresholds.
- Project Report “*Tracking Precipitation in the CoURAGE Domain*” (PDF)

### University of California, Davis

Undergraduate Atmospheric Science Researcher

October 2025 – May 2026

*Advisor: Dr. Matt Igel*

- Designed and automated a parameter sampling workflow in Octave for interacting DoNUT cloud circulations, using physically constrained cloud parameter ranges, duplicate-safe indexing, and systematic output labeling across separation distances from 0-12 km
- Built Python dataframes from model outputs and applied linear, quadratic, and random forest regression to quantify how DoNUT parameters affected extrema
- Validated regression trends by comparing controlled one-parameter-at-a-time simulation sequences against predicted shifts in model response features

### Brookhaven National Laboratory

Atmospheric Science Intern

January 2025 – March 2025

Upton, NY

*Advisor: Dr. Mindy Deng*

- Applied K-means and DBSCAN clustering to XSACR radar observations from the TRACER campaign to isolate features of Houston’s bay breeze

- Developed Python workflows to extract leading edge position and propagation speed, generate time series statistics and compare frontal velocity with overall bay breeze wind speed

## PRESENTATIONS

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**BNL Environmental Science & Technologies Department Intern Seminar**, Upton, NY, August 2026. “Tracking Cold Front Precipitation in the CoURAGE Domain”. Technical presentation to atmospheric science researchers.

**BNL Summer Symposium**, Upton, NY, August 2026. “Tracking Cold Front Precipitation in the CoURAGE Domain”. Selected research presentation for multidisciplinary audience of laboratory interns.

**SSU Annual Math Festival**, Rohnert Park, CA, April 2026. “Unsupervised Machine Learning Techniques to Identify Bay Breeze Statistics” (poster)

## OTHER EXPERIENCE

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|--------------------------------------------------|-----------------------|
| <b>SSU Learning and Academic Resource Center</b> | August 2026 – Present |
| Peer Tutor                                       | Rohnert Park, CA      |

- *Courses Supported:* Physics I & II, Differential Equations, Calculus, Reasoning and Proof, Mathematical Modeling, Probability Theory, Descriptive Astronomy, Finite Math (in-class tutor)

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|----------------------------------|-------------------------|
| <b>Santa Rosa Junior College</b> | January 2026 – May 2026 |
| Finite Math Teaching Assistant   | Santa Rosa, CA          |

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|---------------------------------------|------------------------|
| <b>Brookhaven National Laboratory</b> | May 2025 – August 2025 |
| Summer Policy Student                 | Upton, NY              |

- Utilized prior internship familiarity to support program engagement and ensure a positive experience

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|-------------------------|-------------------------|
| <b>City of Petaluma</b> | June 2024 – August 2024 |
| Water Resources Intern  | Petaluma, CA            |

## TECHNICAL SKILLS

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**Languages:** Python, Octave, C++

**Libraries:** Xarray, Py-ART, scikit-learn, Matplotlib, NumPy, pandas

**Tools:** Jupyter Notebook, NetCDF, L<sup>A</sup>T<sub>E</sub>X, Excel